

Weightlifting Pulling Derivatives: Rationale for Implementation and Application

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Abstract This review article examines previous weightlifting literature and provides a rationale for the use of weightlifting pulling derivatives that eliminate the catch phase for athletes who are not competitive weightlifters. Practitioners should emphasize the completion of the triple extension movement during the second pull phase that is characteristic of weightlifting movements as this is likely to have the greatest transference to athletic performance that is dependent on hip, knee, and ankle extension. The clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull are weightlifting pulling derivatives that can be used in the teaching progression of the full weightlifting movements and are thus less complex with regard to exercise technique. Previous literature suggests that the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull may provide a training stimulus that is as good as, if not better than, weightlifting movements that include the catch phase. Weightlifting pulling derivatives can be implemented throughout the training year, but an emphasis and de-emphasis should be used in order to meet the goals of particular training phases. When implementing weightlifting pulling derivatives, athletes must make a maximum effort, understand that pulling derivatives can be used for both technique work and building strength–power characteristics, and be coached with proper exercise

technique. Future research should consider examining the effect of various loads on kinetic and kinematic characteristics of weightlifting pulling derivatives, training with full weightlifting movements as compared to training with weightlifting pulling derivatives, and how kinetic and kinematic variables vary between derivatives of the snatch.

Key Points

This review article examines previous weightlifting literature and provides a rationale for the use of weightlifting pulling derivatives that eliminate the catch phase for athletes who are not competitive weightlifters.

Previous literature suggests that several weightlifting pulling derivatives may provide a training stimulus for whole body triple extension that is as good as, if not better than, weightlifting movements that include the catch phase.

Practitioners should consider implementing weightlifting pulling derivatives that eliminate the catch phase for athletes who are not competitive weightlifters.

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1 Introduction

Lower body power development is a vital component for an athlete's overall performance in sports that require the completion of the triple extension (hip, knee, and ankle) movement [1–4]. Because the vast majority of sports

require the explosive performance of the triple extension movement (e.g., jumping, sprinting, rapid change of direction), it is of paramount importance that practitioners provide their athletes with methods of training that will allow them to produce the greatest levels of muscular power that transfer to sport performance. Because a large number of training methods that train triple extension exist, selecting the optimal training stimulus may be a trying task.

Previous research has indicated that training with plyometric exercises [5–11], sprints [11–14], whole-body vibration [15–17], and kettlebells [18–20] may improve lower body strength–power characteristics. Despite these training methods that exist for the development and improvement of lower body power, previous research has indicated that weightlifting movements may provide a superior training stimulus [18, 21–24]. As a result, weightlifting movements such as the clean, jerk, snatch, and their derivatives (e.g., power clean, power snatch, etc.) are commonly used to train lower body muscular power via the triple extension movement [3, 4, 25–31]. Weightlifting movements are popular within strength training programs because of the similarities between the triple extension of the lifting movements and those seen in other athletic movements in sports [3]. Specifically, previous research has indicated that strong relationships exist between weightlifting movements and sprinting [32, 33], vertical jump [32, 34, 35], and change of direction [32] ability. The ability of an athlete to accelerate a load (e.g., themselves or an opposing player) and accept a load (e.g., football lineman blocking an opposing lineman) are just some of the training adaptations that can result from training with weightlifting movements. Furthermore, weightlifting movements can emphasize strength in sport specific positions (e.g., baseball player fielding position, ready position for football linebacker, etc.), improve skeletal and soft tissue characteristics [36–41], and also allow the practitioner to easily overload the triple extension movement, ultimately producing superior strength–power characteristics.

It is essential to acknowledge the role of force development in athletic development, with greater maximal strength levels being associated with greater rate of force development (RFD) and power output [42–49]. Baker and Nance [50] demonstrated a strong correlation ($r = 0.79$) between three repetition maximum (3RM) back squat performance and squat jump performance, and even stronger correlations between 3RM back squat performance and 1RM hang power clean performance, in elite rugby league players, with similar findings reported by Nuzzo et al. [45]. Furthermore, Stone et al. [47] observed that stronger athletes (1RM back squat = $2.00 \pm 0.24 \text{ kg} \cdot \text{kg}^{-1}$) generated much higher peak power during

the countermovement jump ($5,079 \pm 2,363$ vs. $3,785 \pm 376 \text{ W}$) and squat jump ($5,464 \pm 2,507$ vs. $3,842 \pm 443 \text{ W}$), when compared with weaker athletes (1RM back squat = $1.21 \pm 0.18 \text{ kg} \cdot \text{kg}^{-1}$). The authors [47] also observed that stronger athletes achieved peak power at 20–40 % 1RM during loaded jumps, compared with weaker athletes, who achieved peak power at 10 % 1RM.

Another aspect of strength training deals with characteristics of RFD. RFD has been shown to markedly affect performance [48]. Data suggest that stronger people produce higher RFD magnitudes [47]. However, one of the problems with training to increase RFD and power through normal strength training modes is that the load has to be decelerated at the end of the range of motion, resulting in an altered force–velocity profile when compared with ballistic exercises where no deceleration is required, such as weightlifting movements (e.g., snatch, clean, and jerk) and their derivatives [4]. In traditional strength training exercises, such as the back squat, this deceleration phase can account for as much as 45 % of the entire range of motion, although this decreases as load increases [51]. Weightlifting derivatives may markedly enhance RFD as the intention is to accelerate throughout the concentric phase [52].

Although less complex exercises can increase lower body power [1, 3, 53], it is clear that previous research supports the use of full weightlifting movements as compared with other training methods for lower body muscular power [18, 21–24]. Sport coaches and some strength and conditioning coaches may have the view that weightlifting movements are injurious to their athletes' wrists and shoulders [54–57], which may be why weightlifting movements are not prescribed often for baseball players [55, 58]. However, this does not mean that athletes cannot benefit from using weightlifting pulling derivatives that remove the catch phase and emphasize the completion of the explosive triple extension movement [54, 56, 59]. In this light, researchers have discussed the technique of weightlifting pulling derivatives [60–63] as well as examined their kinetic and kinematic potential as training exercises [2, 64–68]. Furthermore, weightlifting pulling derivatives have been compared with full weightlifting movements to determine which exercises may produce a superior training stimulus [25, 26, 69, 70]. The purpose of this review article is to examine previous weightlifting literature and to provide a rationale for the use of weightlifting pulling derivatives that eliminate the catch phase for athletes who are not competitive weightlifters. The authors acknowledge that there are many types of weightlifting movement derivatives (e.g., hang power clean, pull to the knee, drop snatch, clean and snatch from the knee, etc.); however, this article discusses weightlifting pulling derivatives that have been more thoroughly

examined within the scientific literature. Furthermore, this article focuses primarily on clean pulling derivatives because of the limited scientific literature that has examined snatch pulling derivatives.

2 Literature Search Methodology

Original and review journal articles were retrieved from electronic searches of PubMed and Medline (EBSCO) databases. Additional searches of Google Scholar and relevant bibliographic hand searches with no limits of language of publication were also completed. **The search strategy included the terms weightlifting, weightlifting derivatives, weightlifting variations, lower body power, power clean, and power snatch.** The last month of the search was December 2014.

3 Previous Weightlifting Literature

Much of the extant weightlifting literature has concentrated on the technique of several different exercises, including the snatch, clean, power clean, hang power clean, jerk, clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull [28, 60–63, 71–97]. The majority of research on the biomechanical aspects of weightlifting has been focused on the snatch exercise [72, 73, 76, 79, 81–91, 93–96, 98]. Some of this literature reports the use of three dimensional kinematics to track the bar bath and examine joint characteristics [76, 84, 90, 95], while other literature reports investigations of the snatch technique of various populations [72, 73, 81–83, 85, 86, 88, 89, 96, 98]. The remaining weightlifting literature has discussed the technique of the clean and power clean [28, 77, 78, 97], hang power clean [77, 80], jerk [71], clean pull and snatch pull [63], hang high pull [61], jump shrug [60], and mid-thigh pull [62] as well as examined technique changes of the clean, power clean, and jerk [87, 92, 98].

A number of previous studies and coaching reviews have sought to identify the “optimal load” for peak power production of the lifter-plus-bar system during weightlifting movements [27, 99–107]. The previous literature indicates that the optimal load for peak power production during the clean, snatch, power clean, and hang power clean occurs at approximately 70 % [99, 102] to 80 % 1RM [27, 100, 101, 103, 107]. However, several studies indicated that no statistically significant differences in power development existed between the loads that produced the greatest power and loads ranging from 60 to 80 % 1RM [99] or 50 to 90 % 1RM [27, 102, 103]. In addition, several Russian coaching reviews have indicated that the optimal load based on the speed, height of the lift,

and rhythm of the movements during weightlifting pulling derivatives such as the clean pull and snatch pull, ranges from 90 to 95 % of the full weightlifting movements [104–106, 108]. Taken collectively, these studies and coaching reviews indicate that the optimal load for peak power production of the lifter-plus-bar system occurs between 70 and 80 % 1RM for the clean, snatch, power clean, and hang power clean exercises, and between 90 and 95 % 1RM for weightlifting pulling derivatives. However, practitioners should be aware that optimal loads beyond the lifter-plus-bar system exist, as several studies indicated that the optimal load for peak power production may also be specific to the barbell [109–111] or can be altered to specific joints [112–115]; thus, the optimal load may change based on the biomechanical approach being used to assess power.

Further weightlifting literature has examined the effect of various loads on kinetic and kinematic variables during weightlifting movements and their derivatives [2, 64, 65, 67–70, 87, 115–117]. It was not the primary purpose of these studies to identify an optimal training load because only a few loads were examined in contrast to the entire loading spectrum (i.e., 0–100 % 1RM). This previous research has examined the snatch [87], clean and jerk [87], power clean [115, 116], hang power clean [69, 70, 117], snatch pull [118], hang high pull [68, 69, 119], jump shrug [64, 69, 70], and mid-thigh pull [2, 65, 67]. The information provided by these studies is crucial for the loading prescriptions of athletes, and thus, it is suggested that future research should continue examining how loads affect the kinetics and kinematics associated with weightlifting movements and their derivatives.

4 Training with Weightlifting Movements

As previously mentioned, weightlifting movements are often implemented because of their similarities to sport activities (e.g., sprinting and jumping) [3]. Moreover, weightlifting movements can be used to increase strength in various positions, such as strength off the floor, at various hang positions, and the mid-thigh position [60–63]. Despite their additional benefits (e.g., improvement in skeletal and soft tissue characteristics [36–41], positional strength, improvement in external load acceptance, etc.), **the primary purpose of training with weightlifting movements is training and overloading the coordinated triple extension movement** [102]. Thus, weightlifting movements should be used for this reason. However, practitioners may also implement weightlifting movements with another purpose in mind. Anecdotal reported purposes including training “the athletic movement of dropping under the bar”, “yielding strength”, and “rapid acceptance of a load” may stray from the primary purpose of the

weightlifting movements, which may lead to incorrect technique during the exercises that may ultimately affect the overall training stimulus. **Let it be noted that this does not mean that weightlifting movements that include the catch phase need to be completely eliminated, but rather, used sparingly, because of the greater injury rates that have been associated with catching** [120, 121]. Indeed there have been arguments that sports like American football use the catch to mimic the impact received during a game. However, the efficacy of this belief has not been investigated.

There is little doubt that training with the full weightlifting movements can result in superior training gains as compared with other training methods [18, 21–24]. **However, it should be noted that in order to receive the greatest benefits from each lift, each lift should be performed properly with an emphasis on completing the second pull phase (i.e., triple extension) with maximum effort.** If full weightlifting movements such as the clean, jerk, and snatch and their power variations (e.g., power clean, power snatch) can provide superior training stimuli as compared with other training methods [18, 21–24], it appears that it would be beneficial for practitioners to prescribe the full weightlifting movements in the resistance training programs for their athletes. However, practitioners should take into consideration that potential negative issues may arise with any training method.

5 “The Catch” Phase

The catch phase of a clean or power clean requires an athlete to drop under the bar, rapidly rotate their elbows around the bar, project their elbows forward, and rack the bar across their shoulders [117]. There are two main problems with this movement. First, the only sport that absolutely requires the catch phase in competitions is weightlifting. An obvious issue with this fact is that the vast majority of athletes that practitioners train are not weightlifters. For example, although the sport of baseball does not require the athletes to drop under a bar and catch a load in a front squat or overhead squat position, weightlifting pulling derivatives may be beneficial for baseball athletes to perform in their training, because of the required completion of the triple extension movement [54, 56, 59]. Because weightlifting movements provide a superior training stimulus for lower body power as compared with other training methods [18, 21–24], practitioners may question if the catch phase is really necessary for all athletes to perform chronically throughout their resistance training programs and if derivatives excluding the catch phase can provide a similar training stimulus. Previous research has indicated that no statistical differences in

force, power, and RFD magnitudes existed between the mid-thigh pull and the mid-thigh power clean performed at the same absolute loads [25, 26]. Trivial effect sizes existed for RFD between the mid-thigh pull and mid-thigh power clean, but small to moderate effect sizes existed for peak force and peak power in favor of the mid-thigh pull. The same authors also indicated that the mid-thigh pull produced statistically greater peak force, peak power, and RFD as compared with the power clean and hang power clean, with large to very large effect sizes being present between exercises [25, 26]. Additional research indicated that the jump shrug and hang high pull produced statistically greater force, velocity, and power as compared with the hang power clean [69]. The results from these studies illustrate that eliminating the catch is not detrimental to the kinetic stimulus of the pulling activity and may actually produce a superior kinetic stimulus when compared with the full lift.

A second issue that arises with the catch phase of the clean or power clean is not completing the second pull phase. Specifically, athletes may not fully extend their hip, knee, and ankle joints during the second pull phase in order to prepare to drop under the bar to perform the catch. The authors, as well as other strength and conditioning coaches, have observed athletes attempting to lift supramaximal loads that cannot be completed with proper lifting technique for a set number of repetitions [122]. While poor coaching is largely responsible, this behavior nevertheless takes place. Unfortunately, this may lead to technique deficiencies that may then carry over into subsequent training sessions [123, 124]. If the primary purpose of weightlifting movements is to train and overload the triple extension movement, practitioners should emphasize the completion of the triple extension movement in order for the greatest transfer to sport movements to occur [125, 126].

6 Weightlifting Pulling Derivatives

Most of the transfer of weightlifting variations to performance comes from the pull, not the catch [127, 128]. This statement ultimately leads to the question: Can the explosive triple extension movement be trained with weightlifting pulling derivatives that do not require the catch phase? Recent literature reflects researchers' interest in this question, as a number of studies have examined several clean and snatch pulling derivatives, including the clean pull [66], snatch pull [118, 129], hang high pull [68, 69, 119], jump shrug [64, 69, 70], and mid-thigh pull [2, 25, 26, 65, 67], discussed their technique [60–63], and discussed their implementation in resistance training programs [54]. It should be noted that each of the previous pulling derivatives could be used as part of the teaching

progression for the full weightlifting movements [60–63, 69]. Thus, less complex exercises could be considered for power development, especially in athletes with limited experience in performing the clean and snatch.

6.1 Clean Pull and Snatch Pull

The first weightlifting pulling derivatives discussed within this review are the clean pull and snatch pull (Figs. 1, 2). DeWeese et al. [63] detailed the technique of clean pull and snatch pull and suggested that both exercises may allow an athlete to become more efficient at producing force with the addition of an overload stimulus. Of the derivatives that will be discussed within this review, the clean pull and snatch pull are the most complex with regard to technique because athletes start from a position coming off the floor. However, large ground reaction forces are produced by the athlete by performing the first and second pulls of the clean [116, 128, 130], with the greatest forces occurring during the second pull phase [128]. In addition, depending upon loading, the clean pull and snatch pull can emphasize strength in number of positions, including off the floor, passing the knee, the transition to the second pull phase,

and the mid-thigh [63, 128]. Furthermore, the finishing position of the clean pull and snatch pull requires full extension at the hip, knee, and ankle joints, but also eliminates additional elevation of the barbell that is typically needed for full weightlifting movements. Unfortunately, limited research exists that has examined the clean pull and snatch pull itself [63, 66, 118, 129].

6.2 Hang High Pull

The hang high pull (Fig. 3) is another weightlifting pulling derivative that eliminates the catch phase that is characteristic of full weightlifting movements. The technique of this derivative was previously described by Suchomel et al. [61]. Similar to the clean pull and snatch pull, the hang high pull emphasizes positional strength at the hang position above the knee, the transition to the second pull phase, and at the mid-thigh position. In order to complete a repetition of the hang high pull, the athlete must complete the triple extension movement and elevate the barbell to chest height [61, 68]. Statistically significant relationships exist between the hang high pull and explosive force production at 50 and 100 ms from the onset of the movement [23], indicating that



Fig. 1 Clean pull sequence



Fig. 2 Snatch pull sequence



Fig. 3 Hang high pull sequence



Fig. 4 Jump shrug sequence

the hang high pull is an explosive movement that could be used to train lower body power. Furthermore, previous studies indicated that the hang high pull produced greater peak force, velocity, and power as compared with the hang power clean performed at the same loads [69]. It should be noted that athletes with less weightlifting experience may sacrifice proper technique to ensure that the bar reaches their chest height. Specifically, an athlete may prematurely “dip” or drop below the bar [61], especially at higher loads. Practitioners should be wary of this common mistake and coach the athlete to fully complete the triple extension movement and adjust the loads as necessary to allow for proper execution of the hang high pull.

6.3 Jump Shrug

A weightlifting pulling derivative that eliminates the elevation of the barbell, but also produces high magnitudes of peak power is the jump shrug [69]. Jump shrug technique has been previously described by Suchomel et al. [60]. Like the previously described weightlifting pulling derivatives, the jump shrug emphasizes strength in the hang position above the knee, the transition to the second pull phase, and the mid-thigh position. The jump shrug is ballistic in nature and requires an athlete to maximally jump as high as possible, resulting in the completion of triple

extension movement in order to leave the ground (Fig. 4) [64]. Previous research has indicated that the jump shrug produced greater peak force, velocity, and power as compared with the hang power clean and hang high pull performed at the same loads [69]. Furthermore, the jump shrug produced greater hip, knee, and ankle joint velocity as compared with the hang power clean performed at several loads [70]. It should be noted that both of the previous studies used Division III National Collegiate Athletic Association (NCAA) track and field athletes and intramural athletes. Practitioners should be aware that no other literature exists on the jump shrug with different populations.

6.4 Mid-Thigh Pull

The strongest and most powerful position during weightlifting movements is the mid-thigh position [97, 116, 131–134]. In order to complete every weightlifting movement with the greatest efficiency, athletes are required to reach the mid-thigh position to begin the second pull phase. Therefore, it should come as no surprise that the mid-thigh pull has been studied as a weightlifting pulling derivative. The technique of the mid-thigh pull has been previously detailed by DeWeese et al. [62]. Due to the importance of reaching the mid-thigh position during all weightlifting movements, being able to perform a pulling

derivative from this position may be beneficial because the starting position is the most important position during the movement. The mid-thigh pull can be performed from an individualized set height on the parallel safety bars of a squat rack (Fig. 5) or from the mid-thigh position outside of the squat rack as a hang variation or from blocks (Fig. 6). The mid-thigh pull is the least complex weightlifting pulling derivative discussed in this review and requires the athlete to rapidly perform triple extension from a set mid-thigh position. Previous research has indicated that the mid-thigh pull produced greater magnitudes of peak force, RFD, and power as compared with the power clean and hang power clean performed at the same absolute loads [25, 26], indicating that this pulling derivative may provide a superior training stimulus as compared with full weightlifting movements that involve the catch phase. However, it should be noted that no statistical differences were found between the mid-thigh pull and the mid-thigh power clean in both of the previous studies.

6.5 Variations of Weightlifting Pulling Derivatives

Weightlifting pulling derivatives can be performed from a variety of positions. The previously mentioned pulling



Fig. 5 Mid-thigh pull sequence from the safety bars of a squat rack



Fig. 6 Sequence of the hang variation of the mid-thigh pull

derivatives have been described to be performed from either the floor (i.e., clean pull and snatch pull) [63], hang position (i.e., hang high pull, jump shrug, and mid-thigh pull) [60–62], and from set heights on the safety bars of a squat rack or blocks (i.e., mid-thigh pull) [62]. What a practitioner must take into consideration is that a pulling derivative performed from one starting position may result in a different stimulus as compared with another. Depending upon the starting position and whether a countermovement or static start is used, a different stimulus may occur. For example, a pulling derivative performed from blocks may provide a different training stimulus as compared with a hang position. A pulling derivative starting from blocks may require greater rates of force production as an athlete must overcome the inertia of the load from a dead-stop position. In addition, a pulling derivative performed from a starting height below the knee will include the double knee bend. Based on the characteristics of an athlete, various training stimuli may need to be provided to develop and further improve specific aspects of performance.

7 Benefits of Weightlifting Pulling Derivatives

Based on statistical differences and effect sizes, previous research supports the notion that less complex weightlifting pulling derivatives may produce greater magnitudes of peak force, RFD, velocity, and power as compared with weightlifting movements that require the completion of the catch phase [25, 26, 69, 70]. However, it should be noted that two of the previous studies [25, 26] indicated that no statistically significant differences existed between the mid-thigh pull and mid-thigh power clean. There are a number of benefits of performing weightlifting pulling derivatives as opposed to the full weightlifting movements. These benefits include evidence indicating that weightlifting pulling derivatives are less complex [60–63], are more time efficient with regard to teaching and learning [28, 135], may potentially decrease the overall impact on the body [54], allow the greater ability to overload a number of fitness characteristics (e.g., peak force, RFD, velocity, and power) [2, 60, 65, 66], and may be used in a variety of set-repetition configurations including strength–power potentiating complexes [129, 136–138].

7.1 Decreased Complexity

Because the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull weightlifting pulling derivatives can be used as part of the teaching progression for the full weightlifting movements [60–63], it should come as no surprise that the pulling derivatives are less complex.

Previous research and a review have indicated that practitioners should consider substituting, at least at times, less complex exercises to train lower body power [53, 69]. By implementing weightlifting pulling derivatives instead of the full lifts, practitioners can eliminate the drop under the bar and catch phases. As previously mentioned, this may eliminate the phases that appear to cause the most technique problems and perhaps the most injuries [120]. Further research has suggested that practitioners should implement pulling derivative exercises to enhance explosive strength during the second pull in less skillful athletes [52]. The amount of experience that novice and intermediate high school and collegiate athletes have with the weightlifting movements varies on the basis of the quality of coaching they received. Instead of implementing the full movements, which may require frequent coaching corrections, it may be more beneficial to implement derivatives that are less complex [52, 69]. Finally, it has been suggested that practitioners should implement exercises that allow acceleration through the entire movement [3, 51]. By implementing exercises of this nature, the training stimulus will mimic specific movement patterns in sports. It is likely that less complex weightlifting pulling derivatives that eliminate the catch phase and focus on the completion of triple extension will encourage athletes to train with movements that demand them to accelerate throughout the entire movement.

7.2 Time Efficient

The second benefit of implementing weightlifting pulling derivatives is that they can be more time efficient to teach and learn as compared with the full weightlifting movements. The full weightlifting movements are more time-consuming to teach and learn as compared with a progression exercise or derivative that does not involve the catch phase [28, 135]. Given the schedules of high school athletes and the current NCAA athlete time restrictions for collegiate athletes, practitioners must maximize quality training time for their athletes. Current NCAA athlete time restrictions allow 20 h per week (4 h per day) during the competitive season and 8 h per week during a non-competitive season. Ultimately, practitioners must consider if it is worth taking the normal training hours allotted to teach the full weightlifting movement that involves the catch. Previous research has indicated that implementing a less complex weightlifting pulling derivative may actually increase quality training time towards lower body strength and power [52, 69]. Instead of using the allotted training time to focus on the technique of a more complex weightlifting movement, it may be more useful to implement a pulling derivative that is less complex.

7.3 Reduced Impact

Training with weightlifting movements typically results in a low injury rate [120, 139, 140]. However, training with full weightlifting movements chronically may increase overuse injuries, especially to the wrists and shoulders of athletes [120, 121]. It is possible that training with weightlifting pulling derivatives may decrease the overall impact on the athlete (i.e., the physical stress placed on the wrists, shoulders, lower back, hips, knees, and ankles required to complete a weightlifting movement). By using weightlifting pulling derivatives, the athlete would decrease the number of collisions with the bar, especially during heavy clean and jerks [120, 121]. It is clear that there may be a need to reduce the number of times an individual turns over the bar. Another benefit of weightlifting pulling derivatives is that the practitioner and athlete may avoid poor full movement technique by eliminating the drop under and catch phases by performing certain weightlifting pulling derivatives from set positions (e.g., mid-thigh pull). Athletes who are highly concerned with the health of their shoulders and wrists, such as baseball players, may benefit from using weightlifting pulling derivatives as opposed to the full lifts [54, 59]. By implementing weightlifting pulling derivatives, the sport coaches may become more open to the idea of exercises that are both highly beneficial in regard to lower body power development and result in decreased stress on the shoulder and wrist joints [54]. This does not necessarily mean that the catch phase has to be completely eliminated, but rather used sparingly. Perhaps a sport such as American football, where catching the weight may provide a stimulus for positive adaptation in impact situations, may benefit from greater use of the catch phase. However, it is previously noted that the efficacy of transfer of training effect from the catch to benefits in on-field impact has not been studied.

7.4 Greater Overload

A key aspect of the specific adaptations to imposed demands (SAID) principle is the overload that the training stimulus provides [141, 142]. When using full weightlifting movements in which the athlete has to perform the catch, the amount of weight a practitioner can prescribe cannot exceed the athlete's 1RM lift. In contrast, another benefit of weightlifting pulling derivatives is that a practitioner can prescribe loads that exceed the 1RM of the athlete [2, 60, 65, 66] because they are not required to catch it. Ultimately, this leads to a greater ability to overload the triple extension movement that is specific to movements in sports. Furthermore, this may lead to specific adaptations such as greater increases in peak force production, RFD,

and power, which are all attributes of a successful athlete. With Comfort et al. [2] reporting that peak force and peak RFD occurred at 120–140 % of 1RM power clean, greater loading will then also allow athletes to express greater power over a wider range of loads using pulling derivatives [143, 144]. The weightlifting pulling derivatives that may be exceptions are the jump shrug and hang high pull. The obvious limitation to the jump shrug is that the athlete is required to land after jumping. If loads in excess of the athlete's 1RM are used in this instance, increased landing forces may result [145]. However, unpublished data on the jump shrug by the authors of this review indicates that as the external load increases, jump height and peak landing forces decrease. With regard to the hang high pull, loading may be limited because of the overall barbell displacement that is required to finish a repetition of the exercise.

7.5 Use in Strength–Power Potentiating Complexes

Previous research has examined weightlifting movements that involve the catch in strength–power potentiating complexes to potentiate the performance of a subsequent high power or high velocity movement [146–151]. Specifically, previous research has examined the potentiating effects of the hang clean [147–149], power clean [150, 151], and power snatch [146]. However, two studies have investigated the potentiating effects of weightlifting pulling derivatives without the catch [129, 137]. Stone et al. [137] investigated the effect of heavy mid-thigh pulls on lighter mid-thigh pulls in international-level weightlifters. The results of their study indicated that peak velocity was statistically enhanced during the potentiation set as compared with the three previous sets. However, peak force, relative peak force, peak power, and RFD did not display statistically significant differences. In another study [129], Chiu and Salem examined vertical jump performance following progressive snatch pulls performed at 70, 80, 90, and 100 % of the subject's 1RM snatch. Their results indicated that the subjects' jump height was increased by 5.77 % at the midpoint of training and 5.90 % at the end of the training session. Further research has indicated that weightlifting pulling derivatives can be implemented in strength training programs as part of a strength–power potentiating complex [136]. Although limited research exists on using weightlifting pulling derivatives as part of strength–power potentiating complexes, practitioners should consider the benefits of weightlifting pulling derivatives above, namely the decreased complexity of the movements and the greater ability to overload the triple extension movement, when designing strength–power potentiating complexes. This notion is supported by a review that discusses the use of ballistic exercises in strength–power potentiating complexes [138]. Because it is

impossible to draw conclusions based on a few studies, it is clear that further research investigating weightlifting pulling derivatives is needed.

8 Implementing Weightlifting Pulling Derivatives

It appears that the weightlifting pulling derivatives discussed above may provide alternatives to weightlifting movements that include the catch phase. The next thing to be considered is how best to implement the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull into resistance training programs. Although limited research exists, several studies have made recommendations on how best to implement these lifts [60–64, 66, 69, 118]. When implementing weightlifting pulling derivatives within resistance training programs, practitioners should consider three main aspects of training: exercise selection, the correct loads to prescribe, and the number of sets and repetitions to prescribe.

8.1 Periodization Model

The weightlifting pulling derivatives discussed above can be implemented throughout entire training macrocycles during different phases of training [60–63]. However, it is obvious that there should be an emphasis and de-emphasis on certain exercises to meet a specific training goal. It is recommended that practitioners should use a phase potentiation (block) model when implementing resistance training programs [152–154]. Combining the concepts from Minetti [155] and Zamparo et al. [156], increases in muscle cross-sectional area and changes in muscle architecture combined with central and local factors including motor unit recruitment, fiber type, and co-contraction enhance the ability to increase maximum strength. Increases in maximum strength combined with central factors, the specificity of the task, and the coordination of multiple joints will then enhance the ability to increase muscular power. Practically speaking, when adopting phase potentiation (block) periodization, the strength–endurance phase should enhance the subsequent strength phase, which should then enhance the strength–power phase of training [152–154]. In this model, the early phases may favor cross-sectional area and strength development, while later phases may emphasize power and velocity. For example, during the accumulation phase of block training (preparation), more emphasis may be placed on higher volumes of pulling movements, with weights ranging from 60 to 110 % of a 1RM clean or snatch, whereas during the transmutation phase, repetitions would decrease and emphasis might shift to power movements using strength–power potentiation complexes. In the competition phase, emphasis may then

be shifted towards cluster training and more complete lifts, for example. The following sub-sections will focus on the individual aspects of the phase potentiation model with regard to weightlifting pulling derivatives.

8.2 Exercise Selection

Exercise selection typically varies with the phase of periodization. Although weightlifting pulling derivatives can be implemented in all phases of training, an emphasis and de-emphasis should be placed on specific exercises to meet the training goals of each phase. When considering the strength–endurance phase, the volume of training is traditionally high with a strong emphasis on exercise technique. Based on the high volume of training, it should be noted that the primary goals of the strength–endurance phase are to increase work capacity and increase muscle cross-sectional area [152, 153]. The clean pull, snatch pull, hang high pull, and mid-thigh pull can all be implemented within the strength–endurance phase. While the athletes may enhance their power–endurance properties during this phase, practitioners should not focus on gaining and improving peak muscular power. Instead the emphasis can be placed on solidifying exercise technique or gaining work capacity for future training blocks with heavier loads. In addition, it should be noted that there may be a lower risk of injury in a fatigued state when performing higher repetitions if an athlete performs additional repetitions of weightlifting pulling derivatives as compared with weightlifting movements that include the catch phase. For example, previous research by Hardee et al. [157] has indicated that when 6 repetitions of the power clean at 80 % 1RM were performed consecutively, exercise technique was altered as compared with performing repetitions with 20 or 40 s of rest in between repetitions. Specifically, the first pull and catch phase were completed in a more forward position during the sixth repetition as compared with the first repetition. Furthermore, there was a decrease in vertical displacement between repetitions 1 and 6. These findings are similar to those of Häkkinen et al. [87], who indicated that snatch and clean and jerk technique failed after 4–6 repetitions. It is crucial to provide several forms of feedback in order to refine exercise technique for heavier repetitions in later phases [91, 158]. Within the strength phase, it is recommended that high force movements are implemented in order to gain positional strength during weightlifting movements, as well as improve peak force production and RFD. For example, the mid-thigh pull could be implemented in a strength phase with loads of ≥ 120 % 1RM power clean to maximize force and RFD [2]. In addition, high force movements may improve peak power and the load range of power development [52, 143]. Finally, during the strength–power phase, the primary goal

is the development of peak power, and therefore a range of loads should be prescribed, as previous research indicates that peak power may be specific to the lifter-plus-bar system [2, 27, 64, 69, 99–103, 117, 119] or barbell [109–111], or can be altered to specific joints [112–115]. Within this phase, a practitioner may consider implementing exercises such as the jump shrug because of its highly ballistic nature at loads of ≤ 45 % 1RM hang power clean to maximize peak velocity and peak power [64, 69].

8.3 General Loading Concepts

Several studies have provided loading information specific for the weightlifting pulling derivatives previously discussed [2, 64–69]. During the strength–endurance phase of training, light to moderately heavy loads (e.g., 60–110 % 1RM) should be prescribed because exercise technique should not be sacrificed. Furthermore, because the volume of training will likely be high, fatigue may alter exercise technique. As a result, it is necessary to keep the external load relatively low to prevent excess fatigue. During the strength phase of training, heavier loads should be prescribed (e.g., ≥ 100 % 1RM) for the development of greater peak force, RFD, and high force power. It should be noted that loads in excess of 100 % 1RM may be used with the clean pull, snatch pull, jump shrug, and mid-thigh pull due to elimination of the catch phase. Previous research has recommended loads near or at the optimal load for each exercise for the development of muscular power during the strength–power phase [159, 160]. However, it should be noted that if practitioners only prescribe loads at or near the optimal load for each exercise, there is a lack of variation in overload stimuli, ultimately preventing the development and improvement of lower body power on all aspects of the force–velocity curve [143]. Therefore, in order to provide optimal training stimuli to their athletes, variations in training loads should be prescribed by practitioners [159, 161]. Based on the order of training phases, loads may be progressively increased or decreased to meet the training goal for each phase, ultimately providing the athlete with the training stimuli to fully develop the force–velocity characteristics [143]. Specific loading recommendations for each weightlifting pulling derivative are provided in Sects. 8.4, 8.5.

8.4 Loading for Peak Force Production

The previous literature suggests that the greatest magnitudes of peak force were produced during the heaviest loads examined during the snatch pull [118], hang high pull [68, 69, 119], and mid-thigh pull [2]. In contrast, Suchomel et al. [64] indicated that the greatest magnitude of peak force during the jump shrug occurred at 65 % of the

subjects' 1RM hang power clean as compared with the heaviest load of 80 % 1RM. However, no statistically significant differences existed. This provides a rationale for training with a range of loads to optimize force production, which is supported by Haff and Nimphius [143]. As previously mentioned, the greatest peak force and RFD during the mid-thigh pull occurs at loads of ≥ 120 % 1RM power clean [2]. As compared with the previous exercises, no information currently exists on the greatest peak forces during the clean pull. However, as Enoka [128] and Souza et al. [116] reported that peak force and RFD occurs during the second pull phase of the clean pull and power clean performed at loads of 70 and 85 % 1RM and 60 and 70 % 1RM, respectively, it is possible that higher forces may occur at heavier loads greater than 80 %. In general, it appears that heavier loads implemented during each weightlifting pulling derivative will allow for the greatest force production, potentially leading to greater peak force production, RFD, and high force peak power and range of peak power adaptations [52, 143]. As mentioned above, the clean pull, snatch pull, jump shrug, and mid-thigh pull derivatives may allow the use of loads in excess of the athletes' 1RM clean or snatch [2, 60, 65, 66, 104–106], thus allowing for the enhanced ability to overload the triple extension movement. In contrast, the maximum load that can be prescribed for the hang high pull is that which corresponds to the 1RM hang power clean of the athlete [61].

8.5 Loading for Peak Power Development

As previously mentioned, training at loads at or near the loads that produce the greatest magnitude of power (i.e., the optimal load) may lead to an improvement in maximal lifter-plus-bar system power [159, 160]. Limited information exists regarding loading information for the clean pull and snatch pull. One study [66] indicated that loads of 90 and 120 % of each athletes' 1RM power clean did not differ in regard to peak power during the clean pull, regardless of the set configuration. However, several Russian coaching reviews indicated that the optimal load based on the speed, height of the lift, and rhythm of the movements during weightlifting pulling derivatives such as the clean pull and snatch pull is approximately 90–95 % of the full weightlifting movements [104–106, 108]. In regard to the hang high pull, previous research has indicated that the greatest peak power occurs between 30 and 45 % of the subjects' 1RM hang power clean [68, 69] or between 30 and 60 % 1RM hang high pull [119] in Division III NCAA male track and field and intramural athletes and Division I NCAA male and female soccer players, respectively. This is in contrast to loads ranging from 70 to 80 % 1RM that have been reported to elicit peak power output during the

power clean and hang power clean [27, 99–103]. Previous research has indicated that the loads that produced the greatest peak power during the jump shrug ranged from 30 to 45 % of the subjects' 1RM hang power clean [64, 69]. While examining the mid-thigh pull, Comfort et al. [2] indicated that the greatest peak power production occurred at 40 % of the subjects' 1RM power clean, but was not statistically different to the peak power that occurred at 60 % 1RM. In contrast, Kawamori et al. [65] reported that peak power production was not statistically different between loads of 30, 60, 90, and 120 % of their subject's 1RM power clean. Although no statistical differences existed, the highest peak power occurred at 60 % 1RM power clean. Moderate and large effect sizes existed between 60 and 90 % 1RM and between 60 and 120 % 1RM, respectively, while a trivial effect size existed between loads of 60 and 30 % 1RM. Haff et al. [67] also examined the peak power produced during the mid-thigh pull at loads of 80, 90, and 100 % of the subject's 1RM power clean and displayed that the greatest peak power occurred at 80 % 1RM. However, it should be noted that the study by Haff et al. [67] only provided descriptive peak power data without a statistical analysis, and thus it is unknown if statistically significant differences exist between the loads examined. In response to the peak power development ranges provided for the weightlifting pulling derivatives, it should be noted that stronger athletes may produce peak power at a higher percentage of 1RM [144]. Thus, heavier relative loads may need to be prescribed for specific athletes that meet this criterion so that they may receive an optimal training stimulus.

As mentioned above, practitioners should be cautious when prescribing "optimal loads," as previous research indicates that loads for peak power development may be specific for the lifter-plus-bar system [2, 27, 64, 69, 99–103, 117, 119] or barbell [109–111], or altered to specific joints [112–115]. Furthermore, the "optimal load" could change with trained state or accumulated fatigue, as determined by relative strength level [47]. This information provides a rationale of why multiple loads should be prescribed during training. In order to effectively implement multiple loads with weightlifting pulling derivatives, coaches should utilize multiple loads through the incorporation of warm-ups, down sets, and heavy versus light days. Additionally, it is likely that fatigue effects could change the optimal load for any measure. By implementing multiple loads, practitioners would provide training stimuli that may lead to a more extensive development of the overall power characteristics of their athletes [143]. Although the vast majority of loading recommendations of weightlifting pulling derivatives are based on the 1RM of a full weightlifting movement, it should be noted that this is not the only method of prescribing loads. Another method

includes prescribing loads based on the best set and repetition schemes completed by the athlete, as detailed by Stone and O'Bryant [162].

8.6 Sets and Repetitions

As previously mentioned, weightlifting pulling derivatives can be implemented in all phases of training, but an emphasis and de-emphasis should be placed on specific exercises to meet the training goals of each phase. In order to realize the fitness characteristics of a specific training phase, the sets and repetitions of that particular phase should be properly structured [152–154]. This concept is no different when implementing weightlifting pulling derivatives such as the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull [60–63]. Previous literature has suggested that blocks of 3×10 (sets $\times$ repetitions) may be implemented for weightlifting pulling derivatives during strength–endurance phases of training to allow for the development of exercise technique and power–endurance attributes [60–63, 163]. As described earlier, the athlete's ability to perform the exercise correctly is of paramount importance and should be considered, especially during a high volume phase in which technique may be negatively affected by fatigue; thus the use of cluster sets may be advantageous [66]. During the strength and strength–power phases of the resistance training program, it has been suggested that practitioners should reduce the volume of weightlifting pulling derivative repetitions (e.g., 3×5 to 3×3), while simultaneously increasing the loads [60–63]. During the strength and strength–power blocks of training, overloading the triple extension movement with high force movements is the primary focus. Because loads are potentially in excess of an athlete's 1RM clean or snatch, limited repetitions can be prescribed to ensure high quality work during each repetition. When implementing weightlifting pulling derivatives during explosive speed and maintenance blocks, the primary goal should be to develop and enhance peak power development. It has been suggested that the volume of training should decrease (e.g., 3×3 , 3×2 , and 2×2), while the loads are also decreased to optimize power output [60–63].

8.7 Set Configurations

The work completed by athletes during resistance training sessions should be of the highest quality in order to realize the greatest transfer to performance. Regardless of the training goal, practitioners should emphasize quality training. When implementing weightlifting movements, this view point should not change. Previous research has indicated that there was an increase in perceived effort

[164], decrease in power output [165], and reduction in exercise technique during consecutive repetitions of the power clean [157], clean and jerk [87], and snatch [87]. However, the use of cluster sets offset the increase in perceived effort [164] and allowed the maintenance of power clean performance [165] and technique [157]. Limited research has investigated the use of cluster sets with weightlifting pulling derivatives. However, a previous study indicated that cluster training sets of the clean pull resulted in statistically greater peak barbell velocity and displacement as compared with a traditional exercise set [66]. Although further research using cluster sets with weightlifting pulling derivatives is needed, evidence using cluster sets with the full weightlifting movements suggests that cluster sets may allow for a decreased perception of effort and maintenance of performance with weightlifting pulling derivatives including the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull [166].

9 Conclusions and Practical Applications

Full weightlifting movements can be highly beneficial if they are performed with proper technique. Practitioners should emphasize the completion of the triple extension movement during the second pull phase that is characteristic of weightlifting movements. Weightlifting pulling derivatives such as the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull may provide a training stimulus that is as good as weightlifting movements that involve the catch phase. Weightlifting pulling derivatives can be implemented throughout the training year, but an emphasis and de-emphasis should be used in order to meet the goals of particular training phases. Three main needs in regard to the athletes exist when implementing weightlifting pulling derivatives. First, the athletes must make a maximal effort, with the intention to accelerate the load as rapidly as possible. Without the appropriate effort, which may require a specific training mentality, it is likely that the transfer of weightlifting pulling derivatives will have reduced effectiveness. Second, the athletes must understand that weightlifting pulling derivatives can be used for both technique work and building strength–power characteristics. As with any exercise, proper technique is the foundation upon which athletes can progress to heavier loads that may ultimately benefit their ability to express strength–power fitness characteristics. Finally, the athletes must be coached appropriately during each weightlifting pulling derivative. As previously mentioned, proper exercise technique is vital; however, the practitioner must take a proactive role with their athletes in providing them with good demonstrations and various forms of feedback [91, 158] in order for the athlete to become an expert with

regard to exercise technique for the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull. In contrast, weightlifters may want to spend more time focusing on the catch phase, as optimal technique during this phase is an essential component of the sport.

9.1 Limitations of Current Research and Recommendations for Future Research

Based on the extant literature and the information provided within this review, it appears that a number of research questions currently exist. Although some literature has examined the effect of various loads on kinetic and kinematic measures during clean pulling derivatives, a paucity of research has examined how different loads affect the kinetic and kinematic potential of snatch pulling derivatives. Minimal research has examined the kinetic and kinematic data of the hip, knee, and ankle joints during weightlifting movements and their pulling derivatives. In fact, only one study has compared the joint work completed during the impact phase of landing from a jump, drop landing, a clean, and a power clean [167]. Although the kinetic data from force plates is largely beneficial to sport scientists, a deeper understanding of weightlifting movements and their pulling derivatives may be attained by examining variables such as joint power and work during the concentric and eccentric phases of each exercise. Research supporting or discounting the catch phase of weightlifting movements is limited, making it difficult to draw conclusions about the purported training benefits of the catch. Finally, no research has compared the training effects between training with weightlifting movements that include the catch and training with weightlifting pulling derivatives that exclude the catch. Future research may consider examining how various loads affect kinetic and kinematic characteristics in order to provide practitioners with the loading information needed to effectively implement weightlifting pulling derivatives. Additional studies that examine kinetic and kinematic joint data are recommended to provide a further understanding of the joint work and power contributing to the end result of weightlifting movements and their derivatives. Training studies comparing the effects of training with full weightlifting movements and weightlifting pulling derivatives must be completed. The information within this review provides the theoretical framework in which weightlifting pulling derivatives can be effective as training stimuli; however, training studies that provide evidence to support various claims are needed. Although this review primarily focused on clean weightlifting movements and their pulling derivatives, limited research has examined various snatch derivatives. Because the snatch exercise requires greater barbell velocity and displacement as

compared with the clean, it is unknown how pulling derivatives of the snatch would affect the kinetics and kinematics of the movement.

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References

1. Baker D, Nance S. The relation between running speed and measures of strength and power in professional rugby league players. *J Strength Cond Res.* 1999;13(3):230–5.
2. Comfort P, Udall R, Jones PA. The effect of loading on kinematic and kinetic variables during the midhigh clean pull. *J Strength Cond Res.* 2012;26(5):1208–14.
3. Hori N, Newton RU, Nosaka K, et al. Weightlifting exercises enhance athletic performance that requires high-load speed strength. *Strength Cond J.* 2005;27(4):50–5.
4. Newton RU, Kraemer WJ. Developing explosive muscular power: implications for a mixed methods training strategy. *Strength Cond J.* 1994;16:20–31.
5. de Villarreal ESS, Requena B, Newton RU. Does plyometric training improve strength performance? A meta-analysis. *J Sci Med Sport.* 2010;13(5):513–22.
6. de Villarreal ESS, Kellis E, Kraemer WJ, et al. Determining variables of plyometric training for improving vertical jump height performance: a meta-analysis. *J Strength Cond Res.* 2009;23(2):495–506.
7. Markovic G. Does plyometric training improve vertical jump height? A meta-analytical review. *Br J Sports Med.* 2007;41(6):349–55.
8. Ebben WP, Feldmann CR, Vanderzanden TL, et al. Periodized plyometric training is effective for women, and performance is not influenced by the length of post-training recovery. *J Strength Cond Res.* 2010;24(1):1–7.
9. Miller MG, Hermiman JJ, Ricard MD, et al. The effects of a 6-week plyometric training program on agility. *J Sports Sci Med.* 2006;5(3):459–65.
10. Ebben WP, Suchomel TJ, Garceau LR. The effect of plyometric training volume on performance. In: Sato K, Sands WA, Mizuguchi S, editors. XXXIInd International Conference on Biomechanics in Sports. Johnson City; 2014. p. 566–9.
11. Rimmer E, Sleivert G. Effects of a plyometrics intervention program on sprint performance. *J Strength Cond Res.* 2000;14(3):295–301.
12. Zafeiridis A, Saraslanidis P, Manou V, et al. The effects of resisted sled-pulling sprint training on acceleration and maximum speed performance. *J Sports Med Phys Fit.* 2005;45(3):284–90.
13. Spinks CD, Murphy AJ, Spinks WL, et al. The effects of resisted sprint training on acceleration performance and kinematics in soccer, rugby union, and Australian football players. *J Strength Cond Res.* 2007;21(1):77–85.
14. MacDougall JD, Hicks AL, MacDonald JR, et al. Muscle performance and enzymatic adaptations to sprint interval training. *J Appl Physiol.* 1998;84(6):2138–42.
15. Marin PJ, Rhea MR. Effects of vibration training on muscle power: a meta-analysis. *J Strength Cond Res.* 2010;24(3):871–8.
16. Cardinale M, Bosco C. The use of vibration as an exercise intervention. *Exerc Sport Sci Rev.* 2003;31(1):3–7.
17. Torvinen S, Kannus P, Sievanen H, et al. Effect of four-month vertical whole body vibration on performance and balance. *Med Sci Sports Exerc.* 2002;34(9):1523–8.

18. Otto WH III, Coburn JW, Brown LE, et al. Effects of weightlifting vs. kettlebell training on vertical jump, strength, and body composition. *J Strength Cond Res.* 2012;26(5):1199–202.
19. Lake JP, Lauder MA. Kettlebell swing training improves maximal and explosive strength. *J Strength Cond Res.* 2012;26(8):2228–33.
20. Manocchia P, Spierer DK, Lufkin AKS, et al. Transference of kettlebell training to strength, power, and endurance. *J Strength Cond Res.* 2013;27(2):477–84.
21. Tricoli V, Lamas L, Carnevale R, et al. Short-term effects on lower-body functional power development: weightlifting vs. vertical jump training programs. *J Strength Cond Res.* 2005;19(2):433–7.
22. Hoffman JR, Cooper J, Wendell M, et al. Comparison of Olympic vs. traditional power lifting training programs in football players. *J Strength Cond Res.* 2004;18(1):129–35.
23. Khamoui AV, Brown LE, Nguyen D, et al. Relationship between force-time and velocity-time characteristics of dynamic and isometric muscle actions. *J Strength Cond Res.* 2011;25(1):198–204.
24. Chiu LZ, Schilling BK. A primer on weightlifting: from sport to sports training. *Strength Cond J.* 2005;27(1):42–8.
25. Comfort P, Allen M, Graham-Smith P. Kinetic comparisons during variations of the power clean. *J Strength Cond Res.* 2011;25(12):3269–73.
26. Comfort P, Allen M, Graham-Smith P. Comparisons of peak ground reaction force and rate of force development during variations of the power clean. *J Strength Cond Res.* 2011;25(5):1235–9.
27. Cormie P, McCaulley GO, Triplett NT, et al. Optimal loading for maximal power output during lower-body resistance exercises. *Med Sci Sports Exerc.* 2007;39(2):340–9.
28. Hedrick A. Teaching the clean. *Strength Cond J.* 2004;26(4):70–2.
29. Hydock D. The weightlifting pull in power development. *Strength Cond J.* 2001;23(1):32–7.
30. Kawamori N, Haff GG. The optimal training load for the development of muscular power. *J Strength Cond Res.* 2004;18(3):675–84.
31. Takano B. The power clean—perspectives and preparation. *Strength Cond J.* 1992;14(1):68–71.
32. Hori N, Newton RU, Andrews WA, et al. Does performance of hang power clean differentiate performance of jumping, sprinting, and changing of direction? *J Strength Cond Res.* 2008;22(2):412–8.
33. Stone MH, Byrd R, Tew J, et al. Relationship between anaerobic power and olympic weightlifting performance. *J Sports Med Phys Fit.* 1980;20(1):99–102.
34. Canavan PK, Garrett GE, Armstrong LE. Kinematic and kinetic relationships between an Olympic-style lift and the vertical jump. *J Strength Cond Res.* 1996;10(2):127–30.
35. Carlock JM, Smith SL, Hartman MJ, et al. The relationship between vertical jump power estimates and weightlifting ability: a field-test approach. *J Strength Cond Res.* 2004;18(3):534–9.
36. Storey A, Smith HK. Unique aspects of competitive weightlifting: performance, training and physiology. *Sports Med.* 2012;42(9):769–90.
37. Gratzke C, Hudelmaier M, Hitzl W, et al. Knee cartilage morphologic characteristics and muscle status of professional weight lifters and sprinters a magnetic resonance imaging study. *Am J Sports Med.* 2007;35(8):1346–53.
38. Grzelak P, Domzalski M, Majos A, et al. Thickening of the knee joint cartilage in elite weightlifters as a potential adaptation mechanism. *Clin Anat.* 2014;27:920–8.
39. Grzelak P, Podgorski M, Stefanczyk L, et al. Hypertrophied cruciate ligament in high performance weightlifters observed in magnetic resonance imaging. *Int Orthop.* 2012;36(8):1715–9.
40. Grzelak P, Polgaj M, Podgórski M, et al. Patellar ligament hypertrophy evaluated by MRI in a group of professional weightlifters. *Folia Morphol.* 2012;71(4):240–4.
41. Heinonen A, Sievänen H, Kannus P, et al. Site-specific skeletal response to long-term weight training seems to be attributable to principal loading modality: a pQCT study of female weightlifters. *Calcif Tissue Int.* 2002;70(6):469–74.
42. Cormie P, McCaulley GO, McBride JM. Power versus strength-power jump squat training: influence on the load-power relationship. *Med Sci Sports Exerc.* 2007;39(6):996–1003.
43. Cormie P, McGuigan MR, Newton RU. Influence of strength on magnitude and mechanisms of adaptation to power training. *Med Sci Sports Exerc.* 2010;42(8):1566–81.
44. Cormie P, McGuigan MR, Newton RU. Adaptations in athletic performance after ballistic power versus strength training. *Med Sci Sports Exerc.* 2010;42(8):1582–98.
45. Nuzzo JL, McBride JM, Cormie P, et al. Relationship between countermovement jump performance and multijoint isometric and dynamic tests of strength. *J Strength Cond Res.* 2008;22(3):699–707.
46. Baker D. A series of studies on the training of high-intensity muscle power in rugby league football players. *J Strength Cond Res.* 2001;15(2):198–209.
47. Stone MH, O'Bryant HS, McCoy L, et al. Power and maximum strength relationships during performance of dynamic and static weighted jumps. *J Strength Cond Res.* 2003;17(1):140–7.
48. Cormie P, McGuigan MR, Newton RU. Developing maximal neuromuscular power: part 2—training considerations for improving maximal power production. *Sports Med.* 2011;41(2):125–46.
49. Baker D, Nance S, Moore M. The load that maximizes the average mechanical power output during jump squats in power-trained athletes. *J Strength Cond Res.* 2001;15(1):92–7.
50. Baker D, Nance S. The relation between strength and power in professional rugby league players. *J Strength Cond Res.* 1999;13(3):224–9.
51. Newton RU, Kraemer WJ, Häkkinen K, et al. Kinematics, kinetics, and muscle activation during explosive upper body movements. *J Appl Biomech.* 1996;12:31–43.
52. Harbili E, Alptekin A. Comparative kinematic analysis of the snatch lifts in elite male adolescent weightlifters. *J Sports Sci Med.* 2014;13(2):417.
53. Janz J, Malone M. Training explosiveness: Weightlifting and beyond. *Strength Cond J.* 2008;30(6):14–22.
54. Suchomel TJ, Sato K. Baseball resistance training: should power clean variations be incorporated? *J Athl Enhanc.* 2013;2(2). doi:10.4172/2324-9080.1000112.
55. Hammer E. Preseason training for college baseball. *Strength Cond J.* 2009;31(2):79–85.
56. Garity B. Effective collegiate baseball strength coaching. *Strength Cond J.* 2009;31(2):74–8.
57. Weatherly J, Schinck C. Concepts for baseball conditioning. *Strength Cond J.* 1996;18(2):32–9.
58. Ebben WP, Hintz MJ, Simenz CJ. Strength and conditioning practices of major league baseball strength and conditioning coaches. *J Strength Cond Res.* 2005;19(3):538–46.
59. Hedrick A. The importance and value of weightlifting movements in baseball. *Natl Strength Cond Assoc Coach.* 2014;1(1):14–5.
60. Suchomel TJ, DeWeese BH, Beckham GK, et al. The jump shrug: a progressive exercise into weightlifting derivatives. *Strength Cond J.* 2014;36(3):43–7.

61. Suchomel TJ, DeWeese BH, Beckham GK, et al. **The hang high pull: a progressive exercise into weightlifting derivatives.** *Strength Cond J.* 2014;36(6):79–83.
62. DeWeese BH, Serrano AJ, Scruggs SK, et al. The midhigh pull: proper application and progressions of a weightlifting movement derivative. *Strength Cond J.* 2013;35(6):54–8.
63. DeWeese BH, Serrano AJ, Scruggs SK, et al. The clean pull and snatch pull: proper technique for weightlifting movement derivatives. *Strength Cond J.* 2012;34(6):82–6.
64. Suchomel TJ, Beckham GK, Wright GA. Lower body kinetics during the jump shrug: impact of load. *J Trainol.* 2013;2:19–22.
65. Kawamori N, Rossi SJ, Justice BD, et al. Peak force and rate of force development during isometric and dynamic mid-thigh clean pulls performed at various intensities. *J Strength Cond Res.* 2006;20(3):483–91.
66. Haff GG, Whitley A, McCoy LB, et al. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J Strength Cond Res.* 2003;17(1):95–103.
67. Haff GG, Stone M, O'Bryant HS, et al. Force-time dependent characteristics of dynamic and isometric muscle actions. *J Strength Cond Res.* 1997;11(4):269–72.
68. Suchomel TJ, Beckham GK, Wright GA. Effect of various loads on the force-time characteristics of the hang high pull. *J Strength Cond Res.* 2014. doi:10.1519/JSC.0000000000000748.
69. Suchomel TJ, Wright GA, Kernozek TW, et al. **Kinetic comparison of the power development between power clean variations.** *J Strength Cond Res.* 2014;28(2):350–60.
70. Suchomel TJ, Wright GA, Lottig J. **Lower extremity joint velocity comparisons during the hang power clean and jump shrug at various loads.** In: Sato K, Sands WA, Mizuguchi S, editors. *XXXIInd International Conference of Biomechanics in Sports.* Johnson City; 2014. p. 749–52.
71. Takano B. Coaching optimal techniques in the snatch and the clean and jerk: Part III. *Strength Cond J.* 1988;10(1):54–9.
72. Schilling BK, Stone MH, O'Bryant HS, et al. Snatch technique of collegiate national level weightlifters. *J Strength Cond Res.* 2002;16(4):551–5.
73. Stone MH, O'Bryant HS, Williams FE, et al. Analysis of bar paths during the snatch in elite male weightlifters. *Strength Cond J.* 1998;20(4):30–8.
74. DeWeese BH, Serrano AJ, Scruggs SK, et al. The pull to knee—proper biomechanics for a weightlifting movement derivative. *Strength Cond J.* 2012;34(4):73–5.
75. DeWeese BH, Scruggs SK. The countermovement shrug. *Strength Cond J.* 2012;34(5):20–3.
76. Gourgoulis V, Aggeloussis N, Antoniou P, et al. Comparative 3-dimensional kinematic analysis of the snatch technique in elite male and female greek weightlifters. *J Strength Cond Res.* 2002;16(3):359–66.
77. Johnson J. Teaching the power clean and the hang power clean. *Strength Cond J.* 1982;4(4):52–4.
78. Takano B. Coaching optimal technique in the snatch and the clean and jerk: part II. *Strength Cond J.* 1987;9(6):52–6.
79. Takano B. Coaching optimal technique in the snatch and the clean and jerk: part I. *Strength Cond J.* 1987;9(5):50–9.
80. Duba J, Kraemer W, Martin G. A 6-step progression model for teaching the hang power clean. *Strength Cond J.* 2007;29(5):26–35.
81. Baumann W, Gross V, Quade K, et al. The snatch technique of world class weight lifters at the 1985 world championships. *Int J Sport Biomech.* 1988;4(1):68–89.
82. Burdett RG. Biomechanics of the snatch technique of highly skilled and skilled weightlifters. *Res Q Exerc Sport.* 1982;53(3):193–7.
83. Isaka T, Okada J, Funato K. Kinematic analysis of the barbell during the snatch movement of elite asian weight lifters. *J Appl Biomech.* 1996;12(4):508–16.
84. Gourgoulis V, Aggeloussis N, Mavromatis G, et al. Three-dimensional kinematic analysis of the snatch of elite Greek weightlifters. *J Sports Sci.* 2000;18(8):643–52.
85. Gourgoulis V, Aggeloussis N, Kalivas V, et al. Snatch lift kinematics and bar energetics in male adolescent and adult weightlifters. *J Sports Med Phys Fit.* 2004;44(2):126–31.
86. Okada J, Iijima K, Fukunaga T, et al. Kinematic analysis of the snatch technique used by Japanese and international female weightlifters at the 2006 junior world championship. *Int J Sport Health Sci.* 2008;6:194–202.
87. Häkkinen K, Kauhanen H, Komi VP. Biomechanical changes in the Olympic weightlifting technique of the snatch and clean and jerk from submaximal to maximal loads. *Scand J Sport Sci.* 1984;6(2):57–66.
88. Frolov VI, Lellikov SI, Efimov NM, et al. Snatch technique of top-class weight-lifters. *Sov Sports Rev.* 1979;14:24–9.
89. Hoover DL, Carlson KM, Christensen BK, et al. Biomechanical analysis of women weightlifters during the snatch. *J Strength Cond Res.* 2006;20(3):627–33.
90. Campos J, Poletaev P, Cuesta A, et al. Kinematical analysis of the snatch in elite male junior weightlifters of different weight categories. *J Strength Cond Res.* 2006;20(4):843–50.
91. Winchester JB, Porter JM, McBride JM. Changes in bar path kinematics and kinetics through use of summary feedback in power snatch training. *J Strength Cond Res.* 2009;23(2):444–54.
92. Winchester JB, Erickson TM, Blaak JB, et al. Changes in bar-path kinematics and kinetics after power-clean training. *J Strength Cond Res.* 2005;19(1):177–83.
93. Gourgoulis V, Aggeloussis N, Garas A, et al. Unsuccessful vs. successful performance in snatch lifts: a kinematic approach. *J Strength Cond Res.* 2009;23(2):486–94.
94. Nejadian SL, Rostami M, Towhidkhan F. Optimization of barbell trajectory during the snatch lift technique by using optimal control theory. *Am J Appl Sci.* 2008;5(5):524–31.
95. Chiu H-T, Wang C-H, Cheng KB. The three-dimensional kinematics of a barbell during the snatch of Taiwanese weightlifters. *J Strength Cond Res.* 2010;24(6):1520–6.
96. Akkus H. Kinematic analysis of the snatch lift with elite female weightlifters during the 2010 world weightlifting championship. *J Strength Cond Res.* 2012;26(4):897–905.
97. Garhammer J. Power clean: kinesiological evaluation. *Strength Cond J.* 1984;6(3):40 (61–3).
98. Kauhanen H, Häkkinen K, Komi PV. A biomechanical analysis of the snatch and clean & jerk techniques of Finnish elite and district level weightlifters. *Scand J Sports Sci.* 1984;6:47–56.
99. Comfort P, Fletcher C, McMahon JJ. Determination of optimal loading during the power clean, in collegiate athletes. *J Strength Cond Res.* 2012;26(11):2970–4.
100. Cormie P, McBride JM, McCaulley GO. The influence of body mass on calculation of power during lower-body resistance exercises. *J Strength Cond Res.* 2007;21(4):1042–9.
101. Cormie P, McBride JM, McCaulley GO. Validation of power measurement techniques in dynamic lower body resistance exercises. *J Appl Biomech.* 2007;23(2):103–18.
102. Kawamori N, Crum AJ, Blumert PA, et al. Influence of different relative intensities on power output during the hang power clean: identification of the optimal load. *J Strength Cond Res.* 2005;19(3):698–708.
103. Kilduff LP, Bevan H, Owen N, et al. Optimal loading for peak power output during the hang power clean in professional rugby players. *Int J Sports Physiol Perform.* 2007;2(3):260–9.

104. Ermakov AD. The training load of weightlifters in pulls and squats. *Weightlifting yearbook*. Livonia: Sportivny Press; 1980. p. 34–8.
105. Frolov VI, Efimov NM, Vanagas MP. The training weights in the snatch pull. *Tyazhelaya Atletika*. Moscow: Fizkultura I Sports; 1977. p. 65–7.
106. Roman RA. The training of the weightlifter. Livonia: Sportivny Press; 1988. p. 39–40.
107. Garhammer J. Weight lifting and training. In: Vaughan CL, editor. *Biomechanics of sport*. Boca Rato: CRC Press Inc.; 1989. p. 169–211.
108. Frolov VI, Efimov NM, Vanagas MP. Training weights for snatch pulls. *Sov Sports Rev*. 1983;18:58–61.
109. Dugan EL, Doyle TL, Humphries B, et al. Determining the optimal load for jump squats: a review of methods and calculations. *J Strength Cond Res*. 2004;18(3):668–74.
110. Hori N, Newton RU, Andrews WA, et al. Comparison of four different methods to measure power output during the hang power clean and the weighted jump squat. *J Strength Cond Res*. 2007;21(2):314–20.
111. McBride JM, Haines TL, Kirby TJ. Effect of loading on peak power of the bar, body, and system during power cleans, squats, and jump squats. *J Sports Sci*. 2011;29(11):1215–21.
112. Moir GL, Gollie JM, Davis SE, et al. The effects of load on system and lower-body joint kinetics during jump squats. *Sports Biomech*. 2012;11(4):492–506.
113. Kipp K, Harris C, Sabick M. Correlations between internal and external power outputs during weightlifting exercise. *J Strength Cond Res*. 2013;27(4):1025–30.
114. Kipp K, Redden J, Sabick M, et al. Kinematic and kinetic synergies of the lower extremities during the pull in Olympic weightlifting. *J Appl Biomech*. 2012;28(3):271–8.
115. Kipp K, Harris C, Sabick MB. Lower extremity biomechanics during weightlifting exercise vary across joint and load. *J Strength Cond Res*. 2011;25(5):1229–34.
116. Souza AL, Shimada SD, Koontz A. Ground reaction forces during the power clean. *J Strength Cond Res*. 2002;16(3):423–7.
117. Suchomel TJ, Beckham GK, Wright GA. The impact of load on lower body performance variables during the hang power clean. *Sports Biomech*. 2014;13(1):87–95.
118. Wicki B, Culici J, DeMarco N, et al. Comparison of rate of force development during a light and moderate load snatch pull. *J Undergrad Kinesiol Res*. 2014;9(2):20–30.
119. Thomas GA, Kraemer WJ, Spiering BA, et al. Maximal power at different percentages of one repetition maximum: influence of resistance and gender. *J Strength Cond Res*. 2007;21(2):336–42.
120. Stone MH, Fry AC, Ritchie M, et al. Injury potential and safety aspects of weightlifting movements. *Strength Cond J*. 1994;16(3):15–21.
121. Kulund DN, Dewey JB, Brubaker CE, et al. Olympic weightlifting injuries. *Phys Sportsmed*. 1978;6(11):111–9.
122. Wagner K, Greener T, Petersen D. Weight room discipline. *Strength Cond J*. 2012;34(2):93–5.
123. Kraemer WJ, Ratamess NA. Fundamentals of resistance training: progression and exercise prescription. *Med Sci Sports Exerc*. 2004;36(4):674–88.
124. Kraemer WJ, Adams K, Cafarelli E, et al. American College of Sports Medicine position stand. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc*. 2002;34(2):364–80.
125. Young WB. Transfer of strength and power training to sports performance. *Int J Sports Physiol Perform*. 2006;1(2):74–83.
126. Verkhoshansky YV, Siff MC. *Supertraining*. Verkhoshansky; 2009.
127. Stone MH. Pulling movements A-Z. In: 8th Annual Coaches and Sport Science College. Johnson City; 2013.
128. Enoka RM. The pull in Olympic weightlifting. *Med Sci Sports*. 1979;11(2):131–7.
129. Chiu LZ, Salem GJ. Potentiation of vertical jump performance during a snatch pull exercise session. *J Appl Biomech*. 2012;28:627–35.
130. Souza AL, Shimada SD. Biomechanical analysis of the knee during the power clean. *J Strength Cond Res*. 2002;16(2):290–7.
131. Beckham GK, Mizuguchi S, Carter C, et al. Relationships of isometric mid-thigh pull variables to weightlifting performance. *J Sports Med Phys Fit*. 2013;53(5):573–81.
132. Kraska JM, Ramsey MW, Haff GG, et al. Relationship between strength characteristics and unweighted and weighted vertical jump height. *Int J Sports Physiol Perform*. 2009;4(4):461–73.
133. Garhammer J. Performance evaluation of Olympic weightlifters. *Med Sci Sports*. 1978;11(3):284–7.
134. Garhammer J. Power production by Olympic weightlifters. *Med Sci Sports Exerc*. 1980;12(1):54–60.
135. Hedrick A, Wada H. Weightlifting movements: do the benefits outweigh the risks? *Strength Cond J*. 2008;30(6):26–35.
136. Baker D. The effectiveness of the wave-cycle for in-season training: 20 years of evidence on the in-season maintenance of strength and power in professional athletes. In: 36th National Strength and Conditioning Association Annual Meeting. Las Vegas; 2013.
137. Stone MH, Sands WA, Pierce KC, et al. Power and power potentiation among strength-power athletes: preliminary study. *Int J Sports Physiol Perform*. 2008;3(1):55–67.
138. Maloney SJ, Turner AN, Fletcher IM. Ballistic exercise as a pre-activation stimulus: a review of the literature and practical applications. *Sports Med*. 2014;44(10):1347–59.
139. Hamill BP. Relative safety of weightlifting and weight training. *J Strength Cond Res*. 1994;8(1):53–7.
140. Pierce K, Byrd R, Stone MH. Youth weightlifting—is it safe? *Weightlifting USA*. 1999;17(4):5.
141. Sale D, MacDougall D. Specificity in strength training: a review for the coach and athlete. *Can J Appl Sport Sci*. 1981;6(2):87–92.
142. Morrissey MC, Harman EA, Johnson MJ. Resistance training modes: specificity and effectiveness. *Med Sci Sports Exerc*. 1995;27(5):648–60.
143. Haff GG, Nimphius S. Training principles for power. *Strength Cond J*. 2012;34(6):2–12.
144. Stone MH, Sands WA, Pierce KC, et al. Relationship of maximum strength to weightlifting performance. *Med Sci Sports Exerc*. 2005;37(6):1037–43.
145. Burkhardt E, Barton B, Garhammer J. Maximal impact and propulsive forces during jumping and explosive lifting exercises. *J Appl Sports Sci Res*. 1990;4(3):107.
146. Radcliffe JC, Radcliffe JL. Effects of different warm-up protocols on peak power output during a single response jump task [Abstract]. *Med Sci Sports Exerc*. 1996;28:S189.
147. Andrews TR, Mackey T, Inkrott TA, et al. Effect of hang cleans or squats paired with countermovement vertical jumps on vertical displacement. *J Strength Cond Res*. 2011;25(9):2448–52.
148. Dinsdale A, Bissas A. Completing a prior set of hang cleans does not improve the performance in the vertical jump irrespective of the length of the recovery period [Abstract]. *J Strength Cond Res*. 2010;24:1.
149. McCann MR, Flanagan SP. The effects of exercise selection and rest interval on postactivation potentiation of vertical jump performance. *J Strength Cond Res*. 2010;24(5):1285–91.
150. Guggenheimer JD, Dickin DC, Reyes GF, et al. The effects of specific preconditioning activities on acute sprint performance. *J Strength Cond Res*. 2009;23(4):1135–9.
151. Seitz LB, Trajano GS, Haff GG. The back squat and the power clean: elicitation of different degrees of potentiation. *Int J Sports Physiol Perform*. 2014;9(4):643–9.

152. Stone MH, Pierce KC, Sands WA, et al. **Weightlifting: program design**. *Strength Cond J*. 2006;28(2):10–7.
153. Bompá TO, Haff G. *Periodization: theory and methodology of training*. Champaign: Human Kinetics; 2009.
154. Stone MH, O'Bryant H, Garhammer J. A hypothetical model for strength training. *J Sports Med Phys Fit*. 1981;21(4):342–51.
155. Minetti AE. On the mechanical power of joint extensions as affected by the change in muscle force (or cross-sectional area), *ceteris paribus*. *Eur J Appl Physiol*. 2002;86(4):363–9.
156. Zamparo P, Minetti A, di Prampero P. Interplay among the changes of muscle strength, cross-sectional area and maximal explosive power: theory and facts. *Eur J Appl Physiol*. 2002;88(3):193–202.
157. Hardee JP, Lawrence MM, Zwetsloot KA, et al. Effect of cluster set configurations on power clean technique. *J Sports Sci*. 2012;31(5):488–96.
158. Rucci JA, Tomporowski PD. Three types of kinematic feedback and the execution of the hang power clean. *J Strength Cond Res*. 2010;24(3):771–8.
159. Kaneko M, Fuchimoto T, Toji H, et al. Training effect of different loads on the force-velocity relationship and mechanical power output in human muscle. *Scand J Sports Sci*. 1983;5(2):50–5.
160. Wilson GJ, Newton RU, Murphy AJ, et al. The optimal training load for the development of dynamic athletic performance. *Med Sci Sports Exerc*. 1993;25(11):1279–86.
161. Moss BM, Refsnes PE, Abildgaard A, et al. Effects of maximal effort strength training with different loads on dynamic strength, cross-sectional area, load-power and load-velocity relationships. *Eur J Appl Physiol Occup Physiol*. 1997;75(3):193–9.
162. Stone MH, O'Bryant HS. *Weight training: a scientific approach*. Minneapolis: Burgess International; 1987.
163. Scala D, McMillan J, Blessing D, et al. Metabolic cost of a preparatory phase of training in weight lifting: a practical observation. *J Strength Cond Res*. 1987;1(3):48–52.
164. Hardee JP, Lawrence MM, Utter AC, et al. Effect of inter-repetition rest on ratings of perceived exertion during multiple sets of the power clean. *Eur J Appl Physiol*. 2012;112(8):3141–7.
165. Hardee JP, Triplett NT, Utter AC, et al. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res*. 2012;26(4):883–9.
166. Haff GG, Hobbs RT, Haff EE, et al. Cluster training: a novel method for introducing training program variation. *Strength Cond J*. 2008;30(1):67–76.
167. Moolyk AN, Carey JP, Chiu LZ. Characteristics of lower extremity work during the impact phase of jumping and weightlifting. *J Strength Cond Res*. 2013;27(12):3225–32.